ECON 5040 Midterm, In Words

For each topic: a short answer to write, using Fisher's variables and one key equation, followed by the math behind it so you actually understand it.

The exam rewards the words; the math box is there so the words make sense to you.

The recipe for every answer

(1) Name the idea. (2) Cite the one key equation and say what each side means. (3) Say what happens when something changes, and why. (4) Sketch the classic graph, if there is one.

1. Inter-temporal prices

How does the real interest rate set the price of consumption today vs. the future? What substitution incentive does a higher rate create?

Sample response (write this)

The gross real rate $R = 1 + r$ is the relative price of c_0 in units of c_1: giving up one unit today buys R units tomorrow, so future consumption costs $1/R$ and the budget line $c_0 + c_1/R = e_0 + e_1/R$ has slope $-R$. At the optimum MRS $= u'(c_0)/\beta u'(c_1) = R$. A higher R makes c_1 cheaper, so the substitution effect always moves the household toward lower c_0 and more saving.

Why it's true (the math)

Where the price comes from. Saving a today gives Ra tomorrow: $c_0 = e_0 - a$, $c_1 = e_1 + Ra$. Eliminating a gives $c_0 + \frac{c_1}{R} = e_0 + \frac{e_1}{R}$, so the prices are $(p_0, p_1) = (1, 1/R)$ and $R = p_0/p_1$.

FOC. Maximize $u(c_0) + \beta u(c_1)$ subject to the budget. The Lagrangian conditions are $u'(c_0) = \lambda$ and $\beta u'(c_1) = \lambda/R$; dividing gives $\frac{u'(c_0)}{\beta u'(c_1)} = R$.

Size of the substitution response (CRRA). With $u'(c) = c^{-\rho}$: $(c_1/c_0)^\rho = \beta R \Rightarrow \frac{c_1}{c_0} = (\beta R)^{1/\rho}$, so $\frac{d\log(c_1/c_0)}{d\log R} = \frac{1}{\rho}$ (the EIS). A higher ρ (more curvature) means a weaker substitution response.

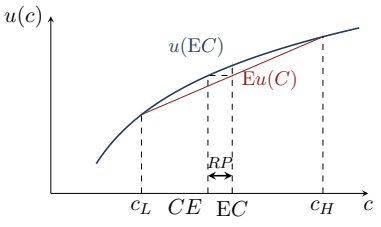
Slutsky caveat. $\partial c_0/\partial R =$ (substitution < 0) + (income effect $\propto (e_0 - c_0)$). For a lender ($e_0 > c_0$) the income effect is positive, so the *total* effect on saving is ambiguous; only the substitution effect is signed.

2. Choice under uncertainty

Expected utility, risk aversion, and the CE of a 50/50 gamble over $\{50, 150\}$ *with* $u = \log c$.

Sample response (write this)

Lotteries are ranked by $\mathrm{E}[u(C)] = \sum_s \pi_s u(c_s)$, and a concave u ($u'' < 0$) means risk aversion: $u(\mathrm{E}C) > \mathrm{E}u(C)$. The certainty equivalent solves $u(CE) = \mathrm{E}u(C)$; here $CE = \sqrt{50 \cdot 150} \approx 86.60 < \mathrm{E}C = 100$, so the risk premium is $RP \approx 13.40$. More curvature gives a lower CE and a higher RP.



Why it's true (the math)

CE computation. $\log CE = \frac{1}{2}\log 50 + \frac{1}{2}\log 150 = \log\sqrt{7500}$, so $CE = 86.60$. In general $CE = u^{-1}(\mathrm{E}u(C))$; for log utility it is the geometric mean, and for CRRA it is $CE = \left(\mathrm{E}[C^{1-\rho}]\right)^{1/(1-\rho)}$.

Jensen. If $u'' < 0$ then $\mathrm{E}u(C) < u(\mathrm{E}C)$, so $u(CE) < u(\mathrm{E}C)$ and, because u is increasing, $CE < \mathrm{E}C$.

Arrow–Pratt (why curvature sets the premium). For a small risk $C = \bar{c} + \varepsilon$ with $\mathrm{E}\varepsilon = 0$ and

Var$\varepsilon = \sigma^2$, Taylor-expand both sides of $u(\bar{c} - RP) = \mathrm{E}u(\bar{c} + \varepsilon)$: $u - u'RP \approx u + \frac{1}{2}u''\sigma^2$, so

$$RP \approx \tfrac{1}{2}\Big(- \frac{u''(\bar{c})}{u'(\bar{c})}\Big)\sigma^2 = \tfrac{1}{2}A(\bar{c})\,\sigma^2, \qquad \rho(c) = cA(c).$$

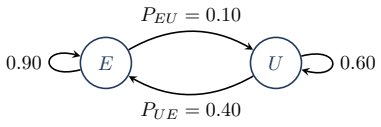
For log, $A = 1/c$ and $\rho = 1$; for CRRA, ρ is constant. Check: $\frac{1}{2} \cdot \frac{1}{100} \cdot 50^2 = 12.5 \approx 13.4$.
vNM uniqueness. Only positive affine transforms $a + bu$ preserve the ranking, since $\mathrm{E}[a + bu] = a + b\mathrm{E}u$.

3. Markov processes

The Markov property; interpret P_{ij} and a row of P.

Sample response (write this)

Markov means $\Pr(X_{t+1} = j \mid X_t = i, X_{t-1}, \dots) = \Pr(X_{t+1} = j \mid X_t = i)$: given today's state, the past adds nothing to the forecast. P_{ij} is the probability of moving from i today to j tomorrow (e.g. $P_{UE} = 0.4$ is the job-finding rate), and row i is the full conditional distribution of tomorrow's state, which sums to 1.



Why it's true (the math)

Distributions. $\psi_{t+1}(j) = \sum_i \psi_t(i)P_{ij}$, i.e. $\psi_{t+1} = \psi_t P$ and $\psi_t = \psi_0 P^t$.
Multi-step (Chapman–Kolmogorov). $(P^2)_{ij} = \sum_k P_{ik}P_{kj}$, summing over intermediate states. With $P = \begin{bmatrix} .9 & .1 \\ .4 & .6 \end{bmatrix}$: $(P^2)_{EU} = .9(.1) + .1(.6) = .15$.
Stationary distribution. $\psi^* = \psi^* P$. Unemployment flows balance: inflow $e \cdot P_{EU}$ equals outflow $u \cdot P_{UE}$, so $.1e = .4u$; with $e + u = 1$ this gives $u^* = \frac{P_{EU}}{P_{EU}+P_{UE}} = \frac{.1}{.5} = 0.2$.
Why rows sum to 1. $\sum_j P_{ij} = \Pr(X_{t+1} \in \text{anything} \mid X_t = i) = 1$. Columns don't have to.

4. Introduction to dynamic programming

Explain the Bellman equation; identify state, choice, current payoff, next state.

Sample response (write this)

The Bellman equation $V(x) = \max_u\{r(x,u) + \beta\mathrm{E}[V(x') \mid x, u]\}$, with $x' \sim g(x,u)$, splits the problem into today's reward plus the discounted expected value of tomorrow's state, assuming optimal behavior from then on. In optimal growth the state is k, the choice is k', the payoff is $\log(Ak^\alpha - k')$, and the next state is k'.

Why it's true (the math)

Why it works (principle of optimality). Separate out the $t = 0$ term of the sequence problem:

$$\max_{\{u_t\}} \mathrm{E}\sum_{t\geq 0} \beta^t r(x_t, u_t) = \max_{u_0}\Big\{ r(x_0, u_0) + \beta\,\mathrm{E}\underbrace{\Big[\max_{\{u_t\}_{t\geq 1}} \mathrm{E}\sum_{t\geq 1} \beta^{t-1} r(x_t, u_t)\Big]}_{=V(x_1)}\Big\}.$$

The tail problem starting at x_1 has the same form, so its value is $V(x_1)$; that's the recursion.
Guess and verify (growth). Guess $V = E + F\log k$. The FOC $\frac{1}{Ak^\alpha - k'} = \frac{\beta F}{k'}$ gives $k' = \frac{\beta F}{1+\beta F}Ak^\alpha$. Matching the $\log k$ coefficients gives $F = \alpha + \alpha\beta F \Rightarrow F = \frac{\alpha}{1-\alpha\beta}$, so $k' = \alpha\beta Ak^\alpha$ and $k^* = (\alpha\beta A)^{1/(1-\alpha)}$.
Euler from Bellman. Combining the FOC $u'(c_t) = \beta V'(k_{t+1})$ with the envelope condition $V'(k) = u'(c)f'(k)$ gives $u'(c_t) = \beta u'(c_{t+1})f'(k_{t+1})$.

5. Value functions and policy functions

Distinguish them; what does each tell you; how is h tied to the Bellman equation?

Sample response (write this)

$V(x)$ is the value of starting in state x and behaving optimally (welfare), while $h(x)$ is the action to take in each state (a contingent plan, $u_t = h(x_t)$). The policy is the argmax of the Bellman equation, $h(x) = \arg\max_u \{r(x,u) + \beta \mathrm{E}[V(x') \mid x, u]\}$, and V is the value of following h forever.

Why it's true (the math)

From V to h: a single maximization in each state. **From h to V_h:** with the policy fixed there's no max, so $V_h = r_h + \beta P_h V_h \Rightarrow V_h = (I - \beta P_h)^{-1} r_h = \sum_{t \geq 0} \beta^t P_h^t r_h$ (a Neumann series: the expected discounted reward stream).

Economic content of V. By the envelope condition $V'(x) = u'(c)$ (cake) or $u'(c)f'(k)$ (growth), so V' is the shadow price of one more unit of the state.

Economic content of h. The law of motion $x_{t+1} \sim g(x_t, h(x_t))$; in a finite DP it induces the Markov chain $P_h(x, \cdot) = M_{h(x)}(x, \cdot)$, whose ψ^* gives long-run outcomes. Examples: $h(k) = \alpha\beta A k^\alpha$; $c^*(x) = (1 - \beta^{1/\rho})x$; McCall's "accept iff $w \geq \bar{w}$."

6. Howard's method vs. value function iteration

The logic of each, and why Howard can be much faster.

Sample response (write this)

VFI guesses V_0 and iterates $V_{n+1} = TV_n$ until V converges, then reads off $h = \arg\max$. Howard guesses a policy h, *evaluates* it exactly with $V_h = (I - \beta P_h)^{-1} r_h$, *improves* it with $\arg\max_u Q(x, u; V_h)$, and stops when h is unchanged. Howard is faster because each evaluation equals infinitely many VFI steps at a fixed policy, so it needs only a few iterations.

Why it's true (the math)

VFI convergence. T is a contraction: $\|TV - TW\|_\infty \leq \beta \|V - W\|_\infty$ (Blackwell: monotonicity plus discounting). So $\|V_n - V^*\| \leq \beta^n \|V_0 - V^*\|$. To cut the error by 10^{-6} you need $n \approx \frac{\ln 10^{-6}}{\ln \beta}$: about 270 steps at $\beta = .95$ and about 1,400 at $\beta = .99$.

Howard convergence. Policy improvement is monotone, $V_{h_{k+1}} \geq V_{h_k}$, and there are at most $|U|^{|X|}$ policies ($2^3 = 8$ for the student problem), so it terminates in finitely many steps, usually just a handful. The trade-off is that each step solves an $|X| \times |X|$ linear system.

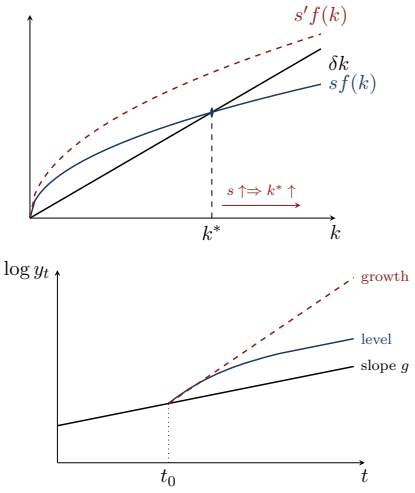
Student check ($\beta = .5$, $h_0 =$ TV): $V_{h_0} \approx (3.64, 6.92, 13.40)$. Sick: TV $1 + .5(.5 \cdot 3.64 + .5 \cdot 6.92) = 3.64$ vs. Study $-1 + .5(.9 \cdot 3.64 + .1 \cdot 6.92) = 0.98$, so keep TV; the same holds in every state, so it stops at once.

7. The Solow model

How saving and capital accumulation pin down k^*, y^*; level vs. growth effect.

Sample response (write this)

Capital per worker evolves as $k_{t+1} = sf(k_t) + (1-\delta)k_t$; diminishing returns make investment eventually just replace depreciation, so the economy converges to $sf(k^*) = \delta k^*$, i.e. $k^* = (sA/\delta)^{1/(1-\alpha)}$. A higher s raises k^* and y^* (a **level effect**), but long-run growth of y per worker equals the rate of technological progress g, so saving doesn't change it.



Why it's true (the math)

Steady state. $sAk^\alpha = \delta k \Rightarrow k^{1-\alpha} = sA/\delta$. Then $\log y^* = \log A + \frac{\alpha}{1-\alpha}\log\frac{sA}{\delta}$, so $\frac{\partial \log y^*}{\partial \log s} = \frac{\alpha}{1-\alpha}$ (about 0.5 with $\alpha = 1/3$). This elasticity is a *level* shift.

Why growth dies out. $\frac{k_{t+1}-k_t}{k_t} = sAk_t^{\alpha-1} - \delta \to 0$ as $k_t \to k^*$, since $k^{\alpha-1} \downarrow$ (diminishing MPK). Any one-time increase in s is eventually absorbed.

Balanced growth. With $E_{t+1} = (1+g)E_t$, $\tilde{k} = K/(EL) \to \tilde{k}^*$, so $Y/L = E\tilde{y}^*$ grows at g and Y grows at $n + g + ng$; s shows up only in $\tilde{k}^*$.

Speed. $\lambda = 1 - \frac{(1-\alpha)(\delta+n+g+ng)}{(1+n)(1+g)} \approx .948$ and half-life $= \frac{\ln .5}{\ln \lambda} \approx 13$ years, independent of s.

Golden rule. $\max_s (1-s)f(k^*)$ gives $f'(k^*) = \delta$, so $s_{GR} = \alpha$.

8. Optimal saving

Interpret $u'(c_t) = \beta R \mathrm{E}_t[u'(c_{t+1})]$; how does the household react when the two sides differ?

Sample response (write this)

The LHS $u'(c_t)$ is the marginal utility of eating one more unit today; the RHS $\beta R \mathrm{E}_t[u'(c_{t+1})]$ is the discounted expected marginal utility of saving it, since it becomes R units tomorrow. If LHS > RHS, the household consumes more and saves less (and the reverse if LHS < RHS) until the two are equal, unless a borrowing constraint binds.

Why it's true (the math)

Perturbation argument. Cut c_t by ε and save it, then raise c_{t+1} by $R\varepsilon$. The utility change is $\Delta \approx \varepsilon\left[-u'(c_t) + \beta R \mathrm{E}_t u'(c_{t+1})\right]$. At an optimum no ε can help, so the bracket is 0. If the bracket is < 0 (LHS > RHS), $\varepsilon < 0$ helps: consume more now.

Why it converges. Raising c_t lowers $u'(c_t)$, and because c_{t+1} falls, $u'(c_{t+1})$ rises ($u'' < 0$). The gap closes.

From Bellman (cake, $R = 1$). The FOC $u'(c) = \beta V'(x-c)$ plus the envelope condition $V'(x) = u'(c)$ give $u'(c_t) = \beta u'(c_{t+1})$. Under CRRA, $c_{t+1}/c_t = (\beta R)^{1/\rho}$; cake eating gives $c^*(x) = (1-\beta^{1/\rho})x$ (sum the geometric series $x = c\sum_j q^j$ with $q = \beta^{1/\rho}$).

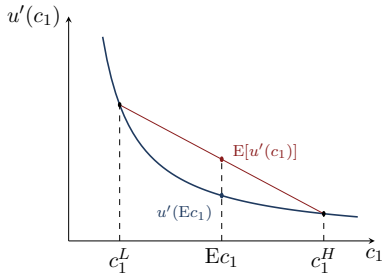
Borrowing limit. The KKT conditions are $u'(c_t) = \beta R \mathrm{E}u'(c_{t+1}) + \lambda$ with $\lambda \geq 0$ and $\lambda(a' - \underline{a}) = 0$, so $u'(c_t) > \beta R \mathrm{E}u'(c_{t+1})$ when the constraint binds.

9. Income fluctuations

Why risk raises saving with $\mathrm{E}Y_1$ *fixed; why assets self-insure.*

Sample response (write this)

From the Euler equation $u'(c_0) = \beta R \mathrm{E}[u'(c_1)]$: if u' is convex ($u''' > 0$), a mean-preserving spread in Y_1 raises $\mathrm{E}[u'(c_1)]$, so the household saves more. This is **precautionary saving**. The asset pays R in every state, but because consumption is chosen after income is realized, the buffer a is drawn down in low-y states, which moves resources to where marginal utility is high (**self-insurance**).



Why it's true (the math)

Jensen on u'. For log utility $u'(c) = 1/c$, which is convex, so $\mathrm{E}\left[\frac{1}{Ra_1+Y_1}\right] > \frac{1}{Ra_1+\mathrm{E}Y_1}$. At any fixed a_1, risk raises the RHS of the Euler equation, so the optimal a_1 must rise to restore equality (raising $u'(c_0)$ and lowering c_1's marginal utility).

Prudence (second order). $\mathrm{E}u'(\bar{c}+\varepsilon) \approx u'(\bar{c}) + \frac{1}{2}u'''(\bar{c})\sigma^2$. The extra term is positive iff $u''' > 0$ (CRRA: $u''' = \rho(\rho+1)c^{-\rho-2} > 0$). Quadratic utility has $u''' = 0$, which means no precautionary saving: *concavity alone isn't enough.*

Example. $y_0 = 10$, $\beta = .95$, $R = 1.02$, $Y_1 \in \{2, 10\}$ vs. a certain 6: the risky case gives a larger a_1.

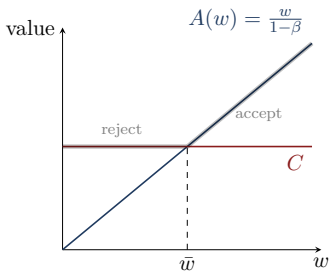
Infinite horizon. $V(a, z) = \max_{a' \geq \underline{a}}\{u(Ra + y(z) - a') + \beta \sum_{z'} P(z, z')V(a', z')\}$. The state is (a, z) because z forecasts y' when income is persistent; it combines DP, a Markov chain, and the Euler equation.

10. Job search (McCall)

Why it is dynamic; state, choice, transition, continuation value; reservation wage.

Sample response (write this)

Rejecting an offer trades today's wage for c plus the option of a new draw $W' \sim F$ tomorrow, so today's choice changes both today's payoff and tomorrow's opportunities. The state is w, the choice is accept or reject, and $V(w) = \max\{\frac{w}{1-\beta}, C\}$ with continuation value $C = c + \beta \mathrm{E}V(W')$. Since $A(w) = \frac{w}{1-\beta}$ rises in w while C is flat (IID offers), the worker accepts iff $w \geq \bar{w} = (1-\beta)C$.



Why it's true (the math)

Accept value. $A(w) = \sum_{t \geq 0} \beta^t w = \frac{w}{1-\beta}$.

Reservation-wage equation. At $\bar{w}$, $\frac{\bar{w}}{1-\beta} = C$, and $V(w') = \max\{\frac{w'}{1-\beta}, \frac{\bar{w}}{1-\beta}\}$. Substituting into C:

$$\frac{\bar{w}}{1-\beta} = c + \frac{\beta}{1-\beta}\mathrm{E}\max\{W', \bar{w}\} \quad \Longrightarrow \quad \bar{w} - c = \frac{\beta}{1-\beta}\mathrm{E}\left[(W' - \bar{w})^+\right].$$

The LHS is the cost of searching one more period (forgoing $\bar{w}$ for c); the RHS is the expected gain from a better draw, valued forever.

Comparative statics from that equation. $c \uparrow \Rightarrow \bar{w} \uparrow$. $\beta \uparrow$ raises the RHS $\Rightarrow \bar{w} \uparrow$. A mean-preserving spread raises $\mathrm{E}(W' - \bar{w})^+$ (a convex function of W') $\Rightarrow \bar{w} \uparrow$: option value.

Computing C. Iterate $C_{n+1} = c + \beta \mathrm{E}\max\{\frac{W'}{1-\beta}, C_n\}$; this is VFI on a scalar.